Spring potential energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate stored spring energy and distinguish it from force.

SUCCESS CRITERIA - I can...

1. Use displacement in meters.
2. Apply the squared relationship for spring energy.
3. Explain how stored energy can become motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “compression” and “spring constant” to compare your model or result. Sentence starters are optional.

compression: shortening or squeezing relative to an unstressed length

spring constant: how much an ideal spring’s force changes per stretch

potential energy: energy associated with the configuration of interacting objects

01 Warm-up: bring back an earlier idea

A cart starts from rest at height 3 m. Can it reach 5 m with no external energy input or losses? Justify using kinetic energy.

Word help: kinetic energy = energy associated with motion

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

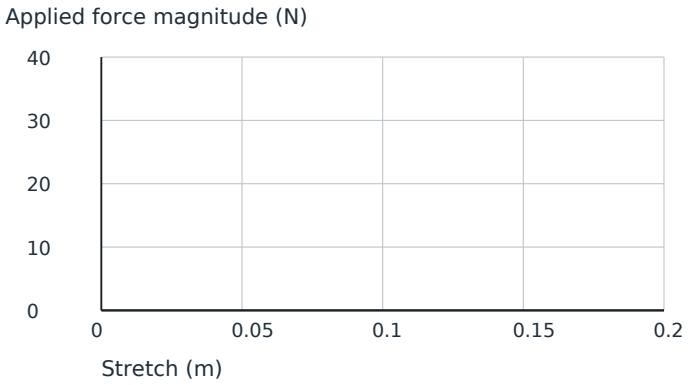
02 Worked example: explain the model

Reference: $U_s = \frac{1}{2}kx^2$; $|F_{spring}| = k|x|$.
s means spring. Compression and stretch both store energy relative to the natural length; energy and force are different quantities.

Worked reasoning: A spring with k=200 N/m stretched 0.10 m stores $\frac{1}{2}kx^2$=1 J. Doubling stretch to 0.20 m stores 4 J, not 2 J.

Your turn: Draw force-versus-stretch triangles and compare their areas.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

The same spring is displaced x and 2 x.
Compare stored energies and force magnitudes.

Sentence starter: Both … . They differ in … because … .



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: ① ② ③

04 Apply the physics

A spring launcher has k = 200 N/m and compression 0.10 m. It launches a 0.50 kg cart along a frictionless horizontal track.
Label the spring model.
Calculate stored energy and the cart's speed when the spring reaches its natural length.

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why is energy positive for both positive and negative x even though spring force reverses?

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 8 N force acts at 120° to a 6 m displacement.
Calculate its work and explain the sign.
Label force and displacement, including their relative direction.

Word help: work = energy transferred by a force along a displacement

Sentence starter: Under the assumption …, the evidence … supports … because … .

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

k=80 N/m; stored energy 0.40 J.
Find displacement magnitude and force magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.5 kg cart is launched by a spring k=400 N/m compressed 0.15 m with no losses.
Find release speed. What compression doubles that speed?

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3